

Sravya Vaddi

Python Developer

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Professional Summary

- Python Developer with 6+ years of experience delivering production-grade backend services, APIs, and ML pipelines across high-stakes industries including healthcare, logistics, and insurance.
- Designed asynchronous APIs using FastAPI and Flask with JWT auth, field-level validation, exception handling, and background workers for high-throughput data systems.
- Architected event-driven systems using Kafka and GCP Pub/Sub, processing over 2M streaming events/day across telemetry and fraud alerting pipelines.
- Migrated legacy monoliths to Kubernetes microservices, improving deployment velocity by 70% and enabling modular, scalable infrastructure on GCP and AWS.
- Delivered ML-backed inference services using PyTorch, Scikit-learn, and FastAPI, powering real-time risk scoring for patient monitoring and claims systems.
- Built Airflow DAGs for batch training, validation, and scoring workflows, aligned with DataFlow and S3 pipelines, improving model refresh cycles by 3x.
- Created CI/CD pipelines with GitHub Actions, Jenkins, and SonarQube, enabling test-first development, secure releases, and rollback-safe blue/green deploys.
- Led async job orchestration with Celery and Redis for non-blocking tasks and notification workflows, reducing frontend load times by over 40%.
- Developed responsive, secure dashboards using ReactJS, Angular, and Tailwind CSS, enabling real-time insights for both internal and client-facing tools.
- Delivered high-reliability APIs with load testing, caching, and failover logic, achieving 99.99% uptime across mission-critical services.
- Owned observability architecture using CloudWatch, ELK, Prometheus, and Grafana; added custom metrics and SLA dashboards for performance triage.
- Maintained 95%+ test coverage across services using Pytest, UnitTest, and Cypress; automated smoke and regression test workflows in CI.
- Implemented OAuth2, token-based auth, and audit logging pipelines to meet HIPAA, GDPR, and SOC2 requirements across regulated systems.
- Mentored junior engineers, reviewed pull requests, led technical walkthroughs, and established service-level design patterns across teams.
- Created internal SDKs and helper libraries to standardize API consumption, reduce code duplication, and accelerate integration velocity.
- Spearheaded documentation strategy including Swagger/OpenAPI specs, Confluence diagrams, and SDK how-tos for 6+ cross-functional teams.
- Defined Agile practices for 3 squads, owning sprint planning, retro facilitation, and backlog prioritization in collaboration with PMs.
- Designed Helm charts and Terraform modules for automated cloud deployments with multi-region scaling, secrets management, and RBAC.
- Created production-ready ETL flows using PySpark and Pandas, driving business-critical dashboards and model pipelines on daily batch data.
- Developed Airflow-triggered retraining and rollback strategies for ML models, integrated with validation checks and model versioning.
- Worked closely with data science and analytics teams to enforce schema validation, error boundaries, and version-safe model exposure.
- Acted as the primary on-call engineer for backend services, resolving live incidents, deploying hotfixes, and authoring RCA documentation.
- Designed Redis-powered rate-limiting and caching layers, improving user API performance and preventing system overload under spikes.
- Led refactoring efforts for technical debt elimination, including modularization, dependency injection, async conversion, and test isolation.
- Built sandbox and staging simulators with seeded test data, allowing QA and product teams to validate edge cases before production rollouts.

- Delivered internal admin portals with real-time analytics, role-based access, and feature toggle control, reducing engineering dependency.
- Enforced GitOps standards across all services — semantic versioning, PR review checklists, CI integration, and automated tagging.
- Presented backend design proposals during quarterly tech councils, including FastAPI best practices, modular DAGs, and Kafka retry logic.
- Designed secure internal services with zero-trust policies, token validation, and scoped permissions, integrating with Okta SSO and API gateways.
- Coordinated cross-team deliverables across frontend, DevOps, QA, and data squads to align platform rollout plans and reduce delivery risk.

Technical Skills

Languages	Python, SQL, JavaScript, TypeScript, Bash, Shell, HTML5, CSS3, C, Java.
Python Frameworks & Libraries	FastAPI, Flask, Django, Pandas, NumPy, PySpark, SQLAlchemy, Pydantic, Marshmallow, Celery, BeautifulSoup, Selenium, Tenacity, asyncio, httpx, BackgroundTasks (FastAPI), Gunicorn, Uvicorn
Web Development	ReactJS, Angular, AngularJS, Redux, Bootstrap, Tailwind CSS, Chart.js, Axios, AJAX, Django Templates, WebSocket, RESTful APIs, Responsive UI, JSON, A11Y (WCAG), Responsive Design, Form Validation
Databases & Storage	PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Aurora PostgreSQL, Redshift, SQL Server, CockroachDB, Avro, Parquet, CSV, JSON
Data Engineering & ETL	Apache Airflow, Kafka, Spark, Hive, GCP Dataflow, AWS Glue, AWS EMR, Dataform, Redshift, Trino, Iceberg, S3 Ingestion, Partitioning, Schema Evolution, Great Expectations, ETL Workflows, Time-series Ingestion, Data Masking
Machine Learning & MLOps	Scikit-learn, TensorFlow, PyTorch, AWS SageMaker, Kubeflow, MLflow, Model Inference, Prediction APIs, Time-series Models, Model Versioning, ML Pipelines, JSON Schema Validation
Cloud Platforms	AWS: Lambda, EC2, S3, SNS, EventBridge, CloudWatch, Glue, IAM, Secret Manager GCP: BigQuery, Pub/Sub, Dataflow, Dataproc, GKE, Cloud Functions, Secret Manager, Dataform
DevOps & Infrastructure	Docker, Docker Compose, Kubernetes, Helm, Helmfile, Terraform, Jenkins, GitHub Actions, Bitbucket Pipelines, Git, GitHub, GitLab, PR Templates, Git Standards, Semantic Versioning, Rollback Automation, LaunchDarkly, Feature Flags, Team Foundation Servers.
Testing & Quality	Pytest, unittest, Coverage.py, Locust, Mock, Postman, Cypress, Test Factories, Linting, Integration Testing, Regression Testing, Git Hooks, Pre-merge Gates
Monitoring & Logging	ELK Stack (Elasticsearch, Logstash, Kibana), Prometheus, Grafana, Grafana Loki, CloudWatch, Sentry, Structured Logging, Trace IDs, Log Rotation, RCA Documentation
Security & Compliance	OAuth2, Auth0, JWT, Scoped API Access, Session Security, Scoped Access Control, Tokenization, GDPR Compliance, HIPAA, Audit Logging, API Gateways
Documentation & Tools	Swagger/OpenAPI, Postman, Confluence, JIRA, Lucidchart, Architecture Diagrams, Developer Wikis, RCAs, Design Docs
Project Methodologies	Agile, Scrum, SDLC, Waterfall, Sprint Planning, Grooming, Retrospectives
Visualization & Analytics	Power BI, Tableau, Matplotlib, Seaborn
Packaging & Environments	Wheel, PEX, Conda, Virtualenv, Visual Studio Code
Certifications	AWS Certified Cloud Practitioner, Salesforce Platform Developer I

Education

M.S., Computer Science | University at Albany, Albany, NY | CGPA : 3.5/4

B.Tech., Electronics and Communications | K L University, India | CGPA : 3.3/4

Professional Experience

Singular Analysts Inc, USA | Python Developer

Feb 2022 – Present

- Led the development of FastAPI-based microservices for alert classification and remediation, integrating AWS Lambda and Step Functions to orchestrate multi-stage processing flows. Enabled sub-second response time under high load in a HIPAA-compliant environment.
- Integrated **Amazon Bedrock** with FastAPI microservices to provide secure, scalable access to foundation models like Claude and Jurassic for generative text summarization and document classification.
- Designed an EventBridge + SNS-driven alert distribution pipeline that reduced critical incident resolution time by 30%, ensuring real-time routing to on-call teams across geographies.
- Refactored legacy Flask endpoints into scalable async FastAPI services using proper ASGI patterns, reducing code complexity and boosting request concurrency by 3x.
- Implemented Uvicorn/Gunicorn hybrid deployments with custom worker tuning, enabling elastic load support and fault-tolerant service behavior in production environments.
- Leveraged asyncio, httpx, and async DB calls to improve latency under traffic spikes, achieving 45% faster response time on telemetry ingestion APIs.
- Developed Celery-based task queues for background processing of ETL jobs and alert evaluation workflows, enabling real-time responsiveness and job retries without blocking the main loop.
- Built Kafka consumers that streamed telemetry from device endpoints into Airflow pipelines, automating alert scoring and flagging anomalies for review by ML systems.
- Wrote reusable Python ETL modules using Pydantic validation and PostgreSQL schema migration support, allowing the team to maintain strong typing and data integrity.
- Set up structured JSON logging with trace IDs, log levels, and rotation support to integrate seamlessly with ELK and Grafana Loki dashboards used by SRE teams.
- Architected a test automation strategy using Pytest, Mocks, and Fixtures with 95%+ coverage across services; fully integrated tests into GitHub Actions workflows with linting and build checks.
- Designed and implemented retry logic using Tenacity and exponential backoff strategies for services sensitive to third-party API failures and transient database issues.
- Tuned PostgreSQL queries by indexing high-cardinality fields, optimizing joins, and partitioning old records to reduce dashboard load latency by over 40%.
- Containerized all microservices using Docker and Helm, implemented RBAC policies for Kubernetes clusters, and collaborated on multi-tenant Helm chart releases with DevOps teams.
- Developed a configuration and secrets manager that supported dynamic overrides, scoped secrets injection, and seamless toggling across dev, staging, and prod environments.
- Created observability dashboards with Grafana using Prometheus exporters, surfacing real-time metrics like task success rates, worker memory usage, and API call latencies.
- Led UI development of a React + Chart.js dashboard that visualized pipeline status, service health trends, and model accuracy metrics consumed by on-call SREs.
- Developed secure Axios-integrated frontend components that pulled data from protected FastAPI endpoints, supporting role-based access control and session token flows.
- Collaborated with DevOps and frontend teams to align build strategies, configure shared CI pipelines, and introduce rollback-safe deployment triggers via GitHub Actions.
- Authored OpenAPI 3.0-compatible Swagger documentation for all endpoints, aligned with internal API governance standards and reviewed during cross-team onboarding.
- Proposed a clean architecture and repository pattern refactor across three core services, reducing module coupling and enabling isolated test coverage per repository.
- Participated in quarterly architectural reviews, providing insights on system bottlenecks, schema design issues, and dependency inversion opportunities.
- Maintained Helm release charts and contributed to multi-region Kubernetes rollout strategies with Helmfile + Terraform provisioning, improving release visibility and safety.
- Delivered mentoring sessions every Friday focused on Python performance, architecture anti-patterns, and writing maintainable async service code for junior team members.
- Defined Git standards and branching policies across teams, introduced PR templates, enforced code style via linters, and promoted test-first contributions with CI gates.

- Owned multiple SDLC checkpoints including story refinement, API spec drafting, service architecture, test automation, CI/CD config, and production rollout monitoring.

Cognizant Technology Solutions, India | **Software Engineer**

Jun 2020 – Oct 2021

- Designed and implemented Python microservices using Flask and FastAPI to handle logistics workflows, including tracking APIs, job dispatching, and delivery validation endpoints.
- Refactored core API logic to follow modular clean architecture, separating concerns via service layers, validators, and dependency-injected database interfaces using SQLAlchemy.
- Developed and productionized Python-based inference APIs for ML models (logistics incident classification), integrating Scikit-learn and Flask for near real-time prediction endpoints.
- Automated backend job queues using Celery and Redis to offload long-running ETL tasks from the main API thread, reducing API response time from 2.5s to under 600ms.
- Authored robust Pytest-based test suites with fixtures, mocks, and parametrized test cases, maintaining 90%+ coverage across all critical backend endpoints.
- Integrated backend services with GitHub Actions-based CI/CD pipelines that enforced linting, coverage checks, and secure secret handling on every pull request.
- Designed PostgreSQL schemas and optimized query execution using EXPLAIN plans and composite indexing for high-traffic endpoints serving 1M+ daily records.
- Built custom logging decorators and exception handlers in Python to generate traceable JSON logs integrated with ELK and Prometheus for observability and debugging.
- Engineered Kafka-based data processors in Python to enrich shipment events with metadata, normalize them, and forward them to downstream data lakes and dashboards.
- Maintained Flask microservices deployed on ECS and Lambda, implementing cold-start mitigation strategies, request caching, and environment-specific secrets handling.
- Developed data validation frameworks using Pydantic and Marshmallow, enforcing schema integrity and logging violations prior to downstream persistence.
- Led backend implementation for multi-step job allocation workflow involving status updates, timeouts, retries, and delivery confirmations managed via stateful APIs.
- Created REST API contracts using OpenAPI/Swagger with proper versioning and backward compatibility, streamlining frontend/backend integration cycles.
- Mentored junior Python developers on best practices for modular coding, async development, API test coverage, and reusable schema validations.
- Containerized backend apps using Docker, defining multi-stage builds, production-ready WSGI configurations (Gunicorn + Flask), and sidecar monitoring.
- Collaborated with ML engineers to wrap time-series predictive models with FastAPI endpoints, handling batch input validation and JSON-to-Numpy transformations.
- Built dynamic configuration systems in Python to manage environment-specific secrets, toggles, and feature flags across staging and production.
- Worked with Airflow to orchestrate backend-triggered batch jobs and ensured successful DAG execution by integrating test hooks and alerting mechanisms.
- Created retry-safe data ingestion layers using idempotency keys, deduplication logic, and transaction-safe PostgreSQL inserts in high-volume ingestion flows.
- Provisioned GKE infrastructure using Terraform and Helm for deploying Python apps, while aligning resource limits with profiling metrics from Prometheus.
- Created custom middleware for Flask apps to handle authentication, authorization, rate-limiting, and user-based context injection into business logic.
- Participated in weekly design discussions, contributing backend architecture diagrams, flowcharts, and trade-off analysis for proposed solutions.
- Reviewed pull requests from backend and data engineering peers, ensuring DRY principles, type safety, and reusable business logic layers.
- Implemented structured exception handling across backend services with user-friendly error messages, logging context, and automated RCA generation tags.
- Owned backend deliverables from sprint estimation to production support, including CI setup, testing, release tagging, rollout validation, and stakeholder demo sessions.

- Developed production-ready Django REST APIs to manage end-to-end insurance workflows, including policy issuance, case routing, audit trails, and claims eligibility validation.
- Migrated legacy Java batch processors into modular Python ETL workflows using Pandas and SQLAlchemy, cutting monthly processing time by over 50% and improving maintainability.
- Designed scalable ETL pipelines using PySpark on AWS EMR to process call transcripts and structured ticket metadata, generating real-time dashboards for claims fraud detection.
- Built cron-based backend scripts in Python to handle alerting, claim expiry notifications, and account reconciliations, triggered via CloudWatch events with retry logic.
- Created parameterized Airflow DAG templates for reusable ingestion pipelines from S3 to Aurora/PostgreSQL and Redshift, supporting dynamic transformation logic per client.
- Refactored internal admin interfaces using Django-Admin, adding scoped role support, audit logging, and field-level masking to comply with GDPR and enterprise compliance policies.
- Integrated Auth0-based OAuth2 authorization into backend services, implementing scoped API access, session lifetimes, and token validation for different business units.
- Documented backend API contracts with OpenAPI and created automated Postman test collections to accelerate integration testing and frontend dev cycles.
- Led data governance enhancements including PII detection, tokenization, and access auditing across all case data APIs and nightly ETL reports.
- Built and managed Elasticsearch pipelines to index customer support logs, enabling full-text search, behavioral analytics, and keyword filtering for audit teams.
- Developed backend endpoints to power AJAX-based dashboards using Django templates and Bootstrap, improving support resolution workflows and UI responsiveness.
- Designed AngularJS widgets and integrated search functionality using Elasticsearch APIs to power dynamic ticket tracking with real-time filters and visual status indicators.
- Optimized long-running PostgreSQL queries by modularizing stored procedures, applying indexing strategies, and adding caching layers, reducing query time from 8s to 1.2s.
- Used Jenkins pipelines to automate Flask-based service testing and Docker deployments, reducing deployment errors across staging and QA by 70%.
- Containerized internal reporting tools using Docker Compose, allowing consistent local environments and faster onboarding for new developers.
- Created structured logging with field masking and correlation IDs across all backend services, enabling centralized analysis through ELK dashboards.
- Partnered with business teams to support data extraction and dashboard onboarding, delivering Power BI views directly backed by API query layers.
- Collaborated with QA teams to define integration test plans, create test data factories in Python, and support cross-environment testing for weekly UAT cycles.
- Wrote Django-based batch workflows with retry safety, validation logs, and modular data loaders, enabling safe replay of failed jobs without data duplication.
- Maintained and updated internal developer wiki documenting ETL job schedules, DAG dependencies, API usage patterns, and team deployment SOPs.
- Used SQLAlchemy and raw SQL execution plans to rewrite inefficient reporting logic, replacing nested loops with window functions and indexed joins.
- Worked closely with infra team to implement alerting via CloudWatch and SNS, monitoring key API health metrics and latency trends across environments.
- Created Selenium + Python scrapers to track competitor product offerings and integrate findings into monthly BI reports and stakeholder dashboards.
- Enhanced UI/UX in Django apps with real-time feedback forms, multi-step forms, and AJAX submission support, improving onboarding experience for policy agents.
- Delivered documentation and training for business analysts, helping them troubleshoot DAG failures, query APIs, and self-serve insurance KPIs via dashboards.